Decision-Making in ABMs

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Gary Polhill

Outline

- What might influence your choice in…
- …your menu of decision-making options
  - Heuristics, Imitation
  - Optimization, Theory
  - Cognitive algorithms, Learning

Modelling context

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- Context of work influences what we do
  - In the social sciences, we are more aware of this
- Contexts in ABMs:
  - Purpose of model and questions you will ask of it
  - Data you have and access to computing resources
  - Computing requirements
  - Expertise, experience and preferences of developer
  - Transparency and comprehensibility to other stakeholders
- Key question:
  - Is this *ever* likely to be *the* model of my case study

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Different Modelling Purposes

PDF

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Context and agents

- How sophisticated an algorithm is really necessary?
- How many choices do they have for each decision?
  - If there's only one option, sophistication is irrelevant
- What are the consequences of the decision for the agent?
  - If consequences are very similar, sophistication is irrelevant
  - If consequences do not 'cascade', sophistication may not be needed
    - The system 'forgets' the choice

Menu of approaches to decision-making

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- Heuristics
  - Simple algorithms that implement ‘rules of thumb’
- Imitation
  - Copying what other agents are doing
    - Descriptive norms per Cialdini et al. (1991)
- Optimization
  - Finding the ‘best’ option
- Using theory
  - Formalizing social theories to build decision-making algorithms
- Cognitive algorithms
  - Drawing on symbolic AI and the cognitive sciences
- Learning
  - Building on experiences

Heuristics

- Advantages
  - Low computational demand for CPU and memory
  - Can be easier to explain to colleagues and stakeholders
  - Can be based on interview evidence
  - Typically fewer parameters
- Disadvantages
  - Sometimes criticized for appearing ‘ad hoc’
  - Agents may be unresponsive to ‘interventions’, or behave strangely in circumstances you had not anticipated

Simple options for heuristics

- Habit – do whatever you last did
  - Theoretical justification – 'personal norm' (Cialdini et al. 1991)
  - You still have to initialize with something though
- Random – just choose an option at random
- Satisficing (Simon 1956) – combine the two
  - One parameter: aspiration for outcome of decision
  - If aspirations met last time, use Habit
  - If not, choose a random option
  - Gotts et al. (2003) showed this was surprisingly effective

Implementing satisficing

```
to-report satisfice [ aspiration result last-choice options ]

  ifelse result >= aspiration [
    report last-choice
  ] [
    report one-of options
  ]

end
```

- aspiration, result, last-choice usually attribute of an agent though
  - No need for the arguments
- options could also be a global, so no need for that argument either
  - Alternatively, a procedure might find out what the options are

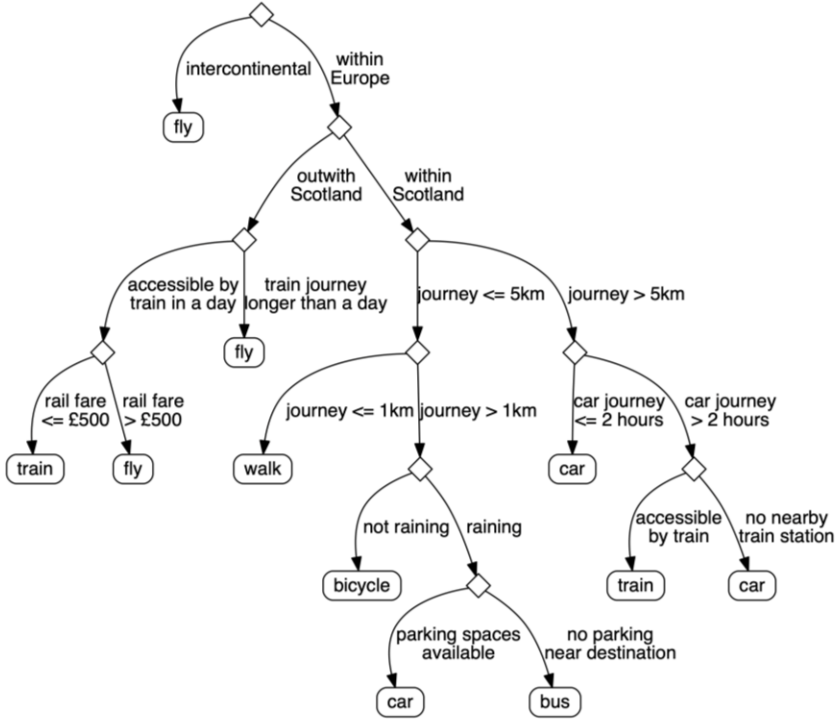
Random-weighted selection

- Each option is given a non-negative score
- Choose an option weighted by that score
  - Create a list of cumulative scores (lines 8-10)
  - Choose a random number in range [0, sum of scores[(line 12)
  - Select the first option with a cumulative weight more than the random number (lines 14-16)

```
2  to-report score [ a-choice ]
3    ; stupid score for demonstration
4    report length (word a-choice)
5  end
6
7  to-report weighted-choice [ choices ]
8    let sum-scores but-first reduce [ [ l i ] ->
9      lput (score i + last l) l
10   ] fput [0] choices
11
12   let r random-float last sum-scores
13
14   report item (first filter [ i ->
15     item i sum-scores > r
16   ] range length sum-scores) choices
17 end
```

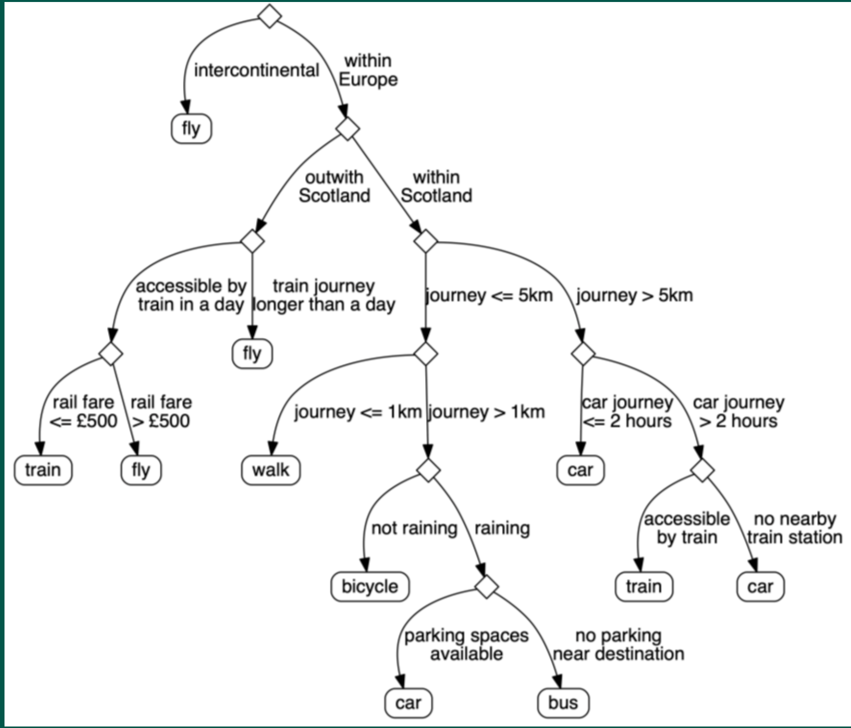
Decision trees

- Easy to implement in NetLogo using ifelse-value nesting
- Machine learning algorithms can construct them from data
  - See next talk
- Easy for people to understand if you want to discuss the algorithm with them
- Can be elicited from stakeholders using interviews
  - Can have different trees for different kinds of people



ifelse-value nesting

```
ifelse-value (intercontinental journey) [
    fly
] [ ; journey within Europe
  ifelse-value (journey beyond Scotland) [
    ifelse-value (train time <= 1 day) [
      ifelse-value (rail fare <= £500) [
        train
      ] [ ; rail fare > £500
        fly
      ]
    ] [ ; train journey > 1 day
      fly
    ]
  ] [ ; journey within Scotland
    ifelse-value (distance <= 5km) [
      ifelse-value (distance <= 1km) [
        walk
      ] [ ; etc.
```



Imitation

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- Advantages
  - Easy to understand
  - Supported by psychological theory (Cialdini et al. 1991)
  - Not too difficult to implement
- Disadvantages
  - Need to be careful how you ask respondents about it
    - Do you copy other people? vs. Do other people copy you?
  - Quite a lot of options for *how* to imitate…

Whom, What and How to imitate

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- Whom to imitate? (Conte & Paolucci 2001)
  - Similarity of attributes
  - Aspirational attributes
  - Similarity of context
  - Imitate the person to imitate!
  - Imitate one person or several?
    - For example, copy the majority
- What to imitate? (Conte & Paolucci 2001)
  - The behaviour or the strategy to choose it?
- How to imitate?
  - Gotts & Polhill (2009) compare algorithms

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Rosaria Conte and Mario Paolucci (2001)

Intelligent Social Learning

Journal of Artificial Societies and Social Simulation vol. 4, no. 1,
<https://www.jasss.org/4/1/3.html>

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Nicholas M. Gotts and J. Gary Polhill (2009)

When and How to Imitate Your Neighbours: Lessons from and for FEARLUS

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<https://www.jasss.org/12/3/2.html>

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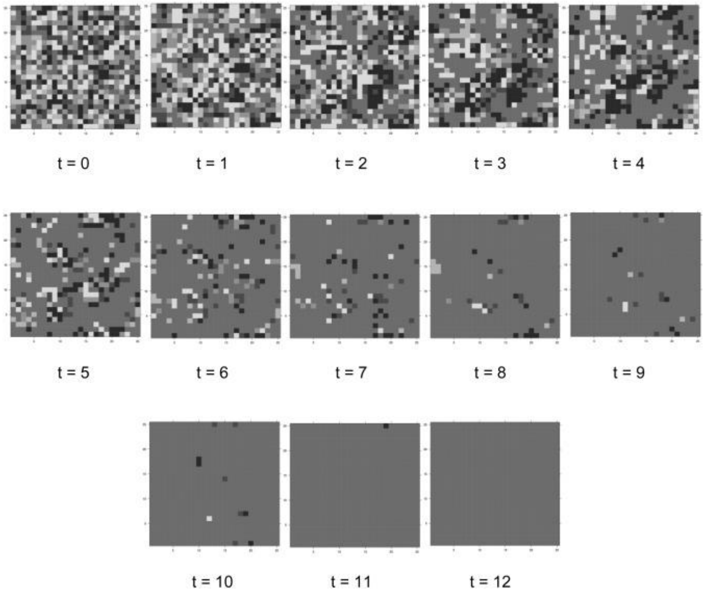
Imitation as a social dilemma

- Trying something new is a risk
  - Experimentation might fail
  - Optimization is costly
- If everyone only copies
  - In the end, everyone does the same thing
  - No available new options to copy
    - See right
- If the environmental and/or social contexts of the decisions change
  - The single option everyone has chosen eventually becomes suboptimal
    - Complex systems are porous…
- Societies need innovators!



(Polhill et al. 2016)

Optimization

- Advantages:
  - Agents make ‘rational’ decisions
    - ‘Boundedly rational’ if information access limited
  - Explainable and credible to economists
- Disadvantages:
  - Optimization is ‘brittle’
    - What was best when you made the decision may no longer be best when you undertake the action and/or experience the consequences
      - Complex systems are porous!
  - Computationally expensive
    - And sometimes infeasible…

Utility maximization

- Implementation
  - Scoring procedure that calculates utility based on preferences (lines 11-16)
    - Basket is an array with one element for each possible item
    - Element contains the amount of that item in the basket
  - Choose which basket you want by comparing them all (lines 18-32)
  - And returning one of the ones with equal maximum utility (line 33)
- Utility maximization is a popular method for decision-making in ABMs
  - Even though criticisms of neoclassical economics are widespread in the ABM community

```
1  globals [
2    options
3  ]
4
5  breed [humans human]
6
7  humans-own [
8    preferences
9  ]
10
11 to-report calculate-utility [ basket ]
12   ; Cobb-Douglas
13   report reduce * (map [
14     [ bask pref ] -> bask ^ pref
15   ] basket preferences)
16 end
17
18 to-report choose-basket [ baskets ]
19   let best-baskets (list first baskets)
20   let best-utility calculate-utility first baskets
21
22   foreach but-first baskets [ basket ->
23     let utility calculate-utility basket
24     (ifelse utility > best-utility [
25       set best-baskets (list basket)
26       set best-utility utility
27     ] utility = best-utility [
28       set best-baskets lput basket best-baskets
29     ] [
30       ; do nothing
31     ])
32   ]
33   report one-of best-baskets
34 end
```

How many baskets…!?

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```
18 to-report choose-basket [ baskets ]
```

- The mammoth in the grotto…
  - We need to generate a list of baskets
- A typical supermarket has at least 10,000 product lines
  - So preferences and each basket is a list of length 10,000
- In one shopping trip you might buy 50 items
  - How many baskets are there to evaluate?
    - Combinations of 10,000 things taken 50 at a time with repetitions
      - See equation below
    - The argument to choose-basket has a length (much (much)) longer than the estimated number of atoms in the universe!

$$\binom{10000 + 50 \ - 1}{50} = \frac{10049!}{50!\,9999!} = \frac{10049 \times 10048 \times \cdots \times 10000}{50!} > \frac{(10^4)^{50}}{3 \times 10^{64}} \approx 3 \times 10^{135}$$

Bounded rationality

- Reduce product lines to explore
  - Only explore products agents have tried
    - Or been advised to try by someone else
  - Allow experimentation with maximum one new one, say
- Use knowledge to restrict choice
  - Why is the agent in the supermarket?
    - If shopping for food, reduce choices to food items
    - Set sensible limits for numbers of each item
- Agents do not know the future
  - If utilities depend on what other agents do, or environmental conditions, you will need to implement functions allowing them to estimate these things

ABM and economics

- Contrasting attitudes to economics in the ABM community
  - "**Economics**, as developed over the past half-century and more, **is not useful** for the analysis and support for formal policy; **it should simply be ignored** by serious social scientists"
    (Moss 1999)
  - A branch of computational economics allowing study of non 'rational' agents (Axtell 2000)
  - "The economy needs agent-based modelling" (Farmer & Foley 2009)
- Key points of departure from economics are based on complex systems thinking
  - Heterogeneity versus single representative agents
  - Interactions and imitation versus ignoring them
  - Heuristic, cognitively plausible, and/or boundedly-rational modes of human decision-making versus full rationality and complete knowledge
    - Moss: focus on the *evidence*. How do you decide what you will buy in the supermarket?
  - Homeostasis and multiple 'basins of attraction' versus equilibria
- No, ABM is not taught in economics undergraduate degrees – they ignore us too!

Complex systems thinking and economics

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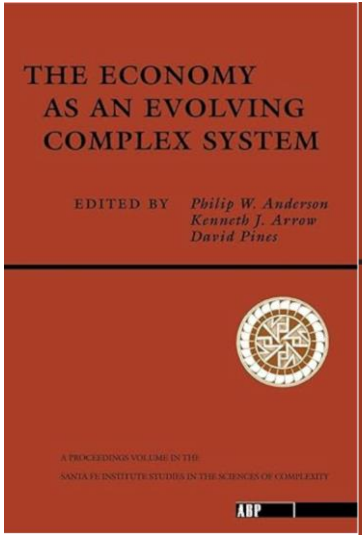
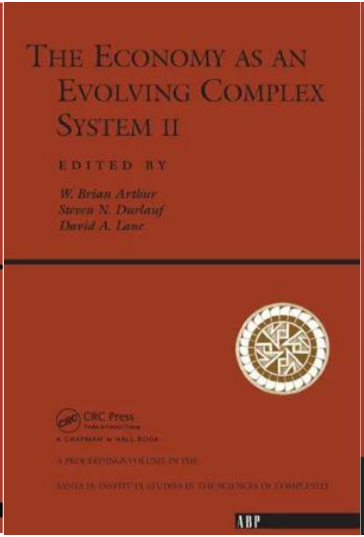
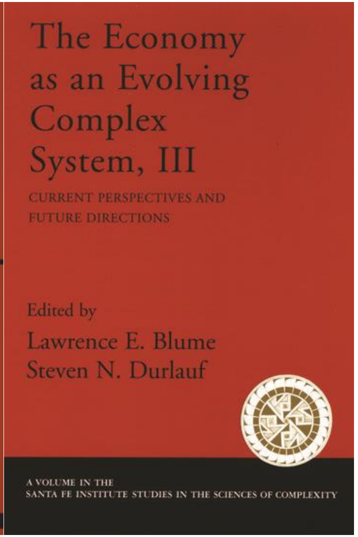
- Not new, but still not mainstream – despite protests about traditional economics!








Economics students call for shakeup of the way their subject is taught

Students from 19 countries argue economics courses failing wider society by ignoring need to address 21st-century issues

Students want courses to include analysis of the financial crash that so many economists failed to see coming. Photograph: Eddie Mulholland/REX

Economics students from 19 countries have joined forces to call for an overhaul of the way their subject is taught, saying the dominance of narrow free-market theories at top universities harms the world's ability to confront challenges such as financial stability and climate change.

1988 | 1997 | 2005 | 2014 | 2026

The Guardian, Nov 2014

Resources on ABM, complex systems and Economics

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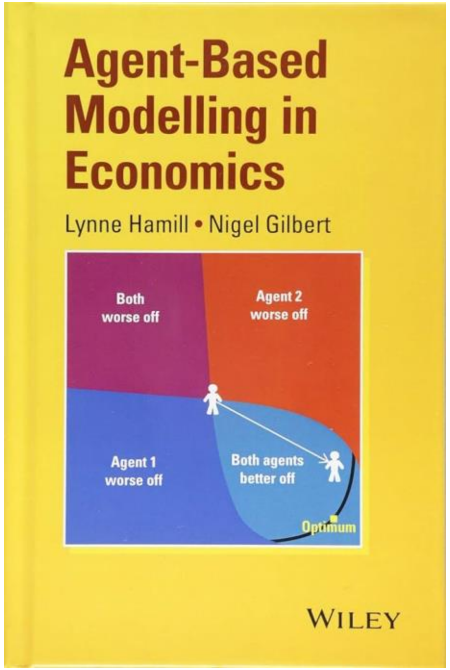
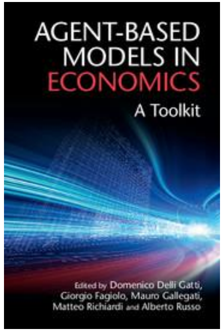
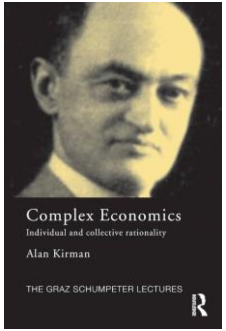
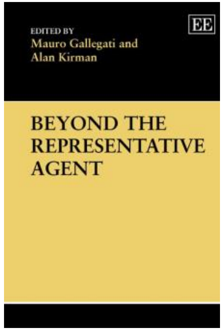
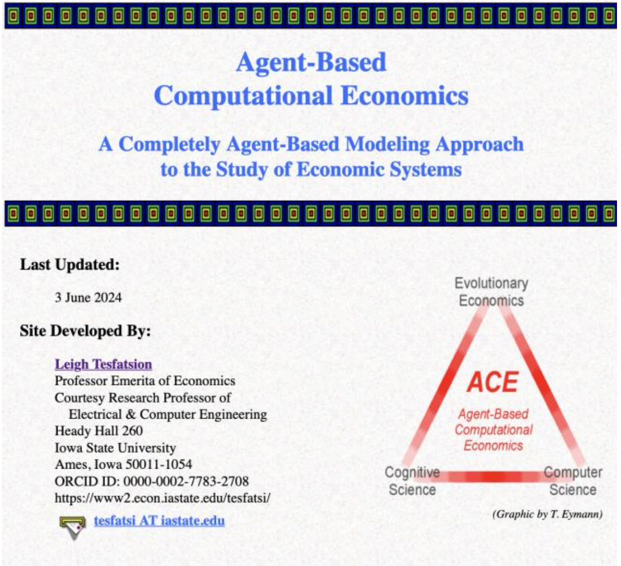








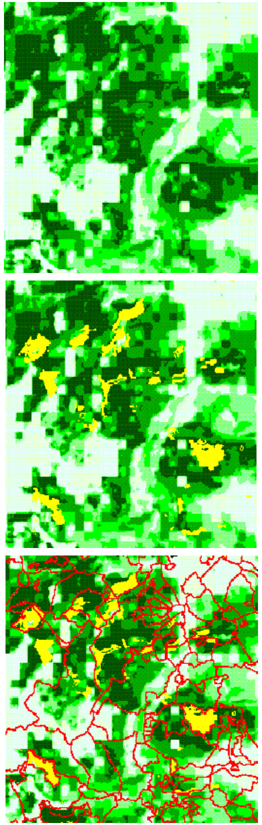


https://faculty.sites.iastate.edu/tesfatsi/archive/tesfatsi/ace.htm

In my own work… Background to FEARLUS

- Farm decisions are not wholly economic
  - In 1995, producers got 24.5p a litre for their milk; the average today is 18p a litre, which represents a loss of more than 3p on every litre. Kemble Farms has been getting 19p a litre.
  - The irony for Colin Rank, one of the family that owns Kemble Farms, is that his cows drink water from a Cotswold spring that he could bottle and sell for 80p a litre
  - “We’re giving it to cows and devaluing it by turning it into milk. Like all dairy farmers we could pack up tomorrow and do something better with our capital, but we do it because we have an emotional investment in the land and the animals.”

(The Guardian, 24 April 2007)



High P(forestry) based on land suitability

Actual forestry

Actual forestry with ownership boundaries

Somewhere in Scotland, mid-1990s
(Aspinall & Burnie, unpub.)

Drawing on Social Theory

- Advantages:
  - Unlike heuristics, theory-based implementations *look* like they are less ad hoc
  - A good way of demonstrating relevance to the social sciences and showing that you are acknowledging the work done there
- Disadvantages:
  - Social theories are not specifications for algorithms
    - You will still have to make stuff up!
  - In the (admittedly unlikely) event that they encounter your work, the proponents of the theories may not be happy with what you've done
    - Though (ideally) this could lead to theoretical advances, disambiguation, and discussion about a theory

Theories aren't specifications

- Psychological theory is more commonly formalized than sociological
  - Can be a bit 'thin' on interactions – especially Western psychology
- Psychologists are more willing to entertain formalization
- Even so, multiple people reading the same theory can implement them in different ways
  - The discipline of having to implement something in code can show what is missing
- Theories in the social sciences are fluid
  - They are refined and updated despite having the same name
  - It can be difficult to pin down what the theory actually is
- Theories in the social sciences are incomplete
  - They may specify one aspect of a system, but you need more (or other theories) to create a workable model

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One Theory - Many Formalizations: Testing Different Code Implementations of the Theory of Planned Behaviour in Energy Agent-Based Models

Hannah Muelder[a] and Tatiana Filatova[b]

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Other articles by these authors

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Example: Consumat (Jager 2000)

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- Context-sensitive decision-making
- Multiple theories:
  - Needs (Max-Neef 1992)
  - Human-env change (Vlek 2000)
  - Behav control (Ajzen & Madden 1986)
  - Operant conditioning (Skinner 1938)
  - Theory of Planned Behaviour (Ajzen 1991)
  - Social Learning (Bandura 1977)
  - Social comparison (Festinger 1954)
- Important (unanswered/explored) questions:
  - Are these theories mutually compatible?
  - Do they have consistent assumptions?
  - How would the theorists feel about their work being combined in this way?

What is the context of the decision?

		Satisfaction of needs	
		High	**Low**
Uncertainty	**High**	Do what most others do	Do what others most like me do
	Low	Habit	Optimize

Cognitive Algorithms

- Advantages
  - A long history of formalizing psychological theory in Symbolic Artificial Intelligence
    - Boden (1988) *Computer Models of Mind*
  - Distributed (Symbolic) AI was one of the many routes through which people encountered ABM
- Disadvantages
  - Algorithms can be computationally demanding
  - Symbolic AI systems are 'brittle' to novelty

Examples

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- Beliefs-Desires Indentions (BDI)
  - Based on logics derived from Dennett's (1981) 'Intentional Stance'
- Rule-based systems (aka 'production systems')
  - Database of knowledge or 'facts'
  - Set of rules that match facts and can be used to derive decisions
- Expert systems
  - Rule-based systems with explanations
    - Which rules and facts were used to make a decision

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Tina Balke and Nigel Gilbert (2014)

University of Surrey, United Kingdom

How Do Agents Make Decisions? A Survey

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Case-Based Reasoning

- Advantages
  - AI ‘pedigree’
  - Evidence-based
  - Adaptive
- Disadvantages
  - Doesn’t tell you everything you need to make a decision

Brief history of Case Based Reasoning

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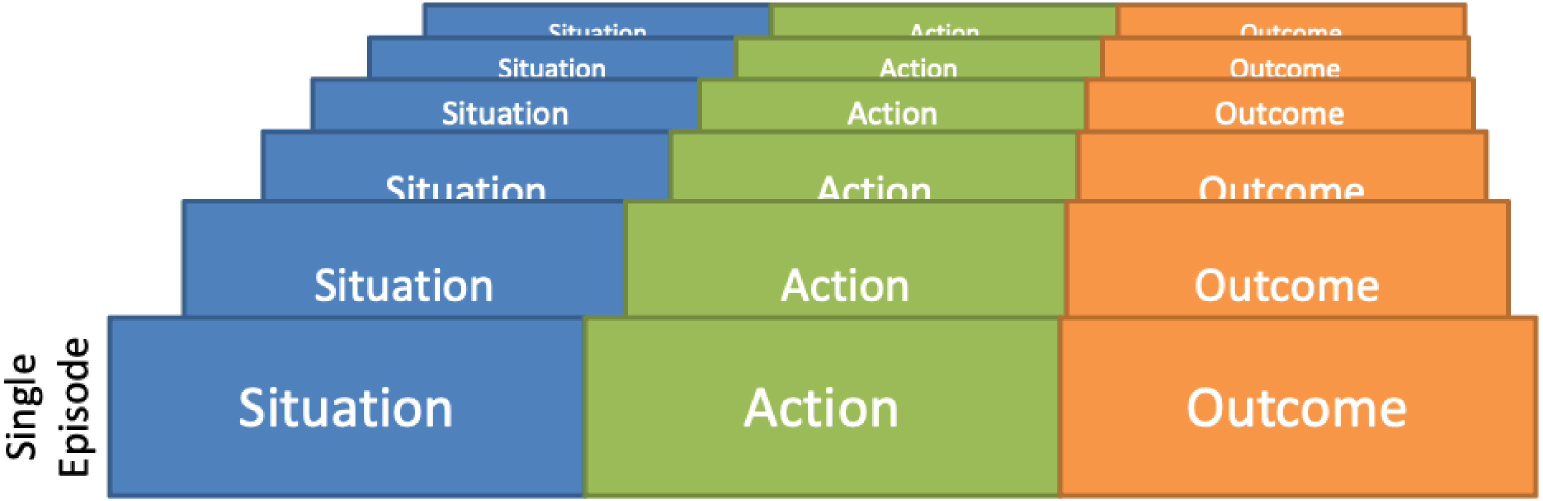
- See Izquierdo et al. (2004) section 3
- Case-based reasoning is a 'GOFAI' algorithm
  - Developed by Aamodt & Plaza (1994)
  - History in cognitive science
    - e.g. Schank & Abelson 1977
  - Based on the way experts use their knowledge and experience to make decisions in new situations
  - Found to have some support in the psychology literature as a means by which humans problem solve (Ross 1989)
- Applications in industry (e.g. Watson 1997)
- Even some consideration by economists
  - e.g. Gilboa & Schmeidler (1995)

Episodic Memory



Data structure with 'relevant' attributes:

- Social
- Environmental
- Individual

Information about chosen option

- Could be a simple ID
- Could be a complex plan or series of actions

Again, 'relevant' data

- Potential issue of attribution
  - Was this really because of the action?
- Comparison operator needed
  - Multidimensional outcomes lead to 'trade-offs'

Making a decision with CBR

- Use the episodic memory
  - Build a data structure that describes the current or expected context
  - Search the episodic memory for similar cases
  - For each different decision in the similar cases
    - Evaluate the distribution of outcomes
- Missing information:
  - How to choose which decision to make!
    - Could use agent attributes (e.g. risk aversion) to choose
    - Could use heuristics
  - What to do if nothing matches or none of the outcomes are good enough
    - Could try something new (experimentation)
    - Could look at what other agents do in analogous situations (descriptive norms)
    - Could pass the expected context to another agent and ask them what to do (advice)

‘Proper’ versus ‘Simple’ CBR

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- As a symbolic AI algorithm, proper CBR:
  - Uses sophisticated analogical reasoning and knowledge bases
  - Can be based on semantic structure and abstraction of situations
  - Less likely to be feasible for large numbers of agents
- Simple CBR
  - Replaces ‘match’ with ‘equals’
  - No sophisticated analogical reasoning or abstraction
  - Feasible in ABMs

Learning and adaptation

- Advantages
  - Common use-case for ABMs entails prodding the system with an intervention to see if it achieves a policy goal
  - Agents will not continue to make decisions robotically even if the decisions are ineffective
- Disadvantages
  - Very little evidence on how real humans adapt decision-making processes
  - Tuning learning parameters can be a challenge
  - Are the agents learning to ‘play’ your model rather than adapting to a phenomenon of real-world interest?

Adaptation

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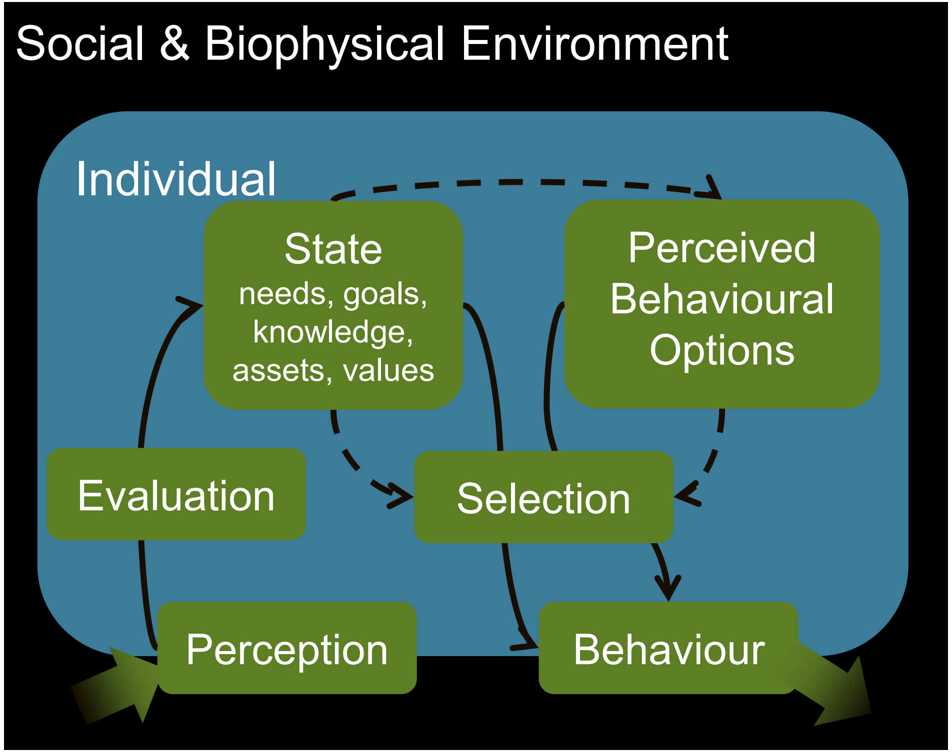
- Population-level 'learning'
- Agents have fixed attributes that affect decision-making
- Selective pressure applied
  - Agents that make decisions having poor outcomes are removed
  - Agents with more successful attributes replicate
  - New agents created to explore other attribute values
- No need to change the decision-making algorithm
- Need some measure of success or fitness
  - Uncomfortable Social Darwinist connotations notwithstanding…
- Risk that 'success' depends on particular attributes of other agents that happen to be in the population at critical points in the simulation run
  - But that's *interesting* from a complexity perspective, and a potential refutation (if any were needed) of Social Darwinism

Learning

- Strictly, agents *learn* when they adjust their attributes and/or the algorithm itself to make a better decision next time they are in a similar situation
  - Happens with Case Based Reasoning when each decision and its outcome is added to the episodic memory
- Can be used with heuristic algorithms
  - Habit-Random satisficing:
    - If outcome >= aspiration, do what you did last time
    - Else, try something new
  - Adjust the aspiration
    - Increase aspiration if someone else made a different choice and had a better outcome
    - Decrease aspiration if someone else chose what agent would have done habitually, and it was better than the outcome from the random experimentation the agent did
  - But… by how much should the aspiration change?
    - And how should it change (e.g. linearly or nonlinearly)…?
  - These choices have consequences for model behaviour

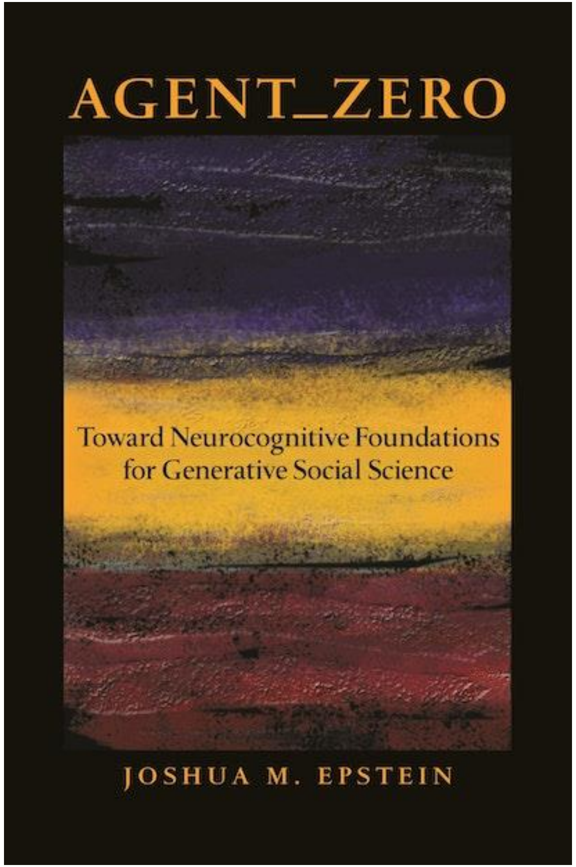
MoHuB: A framework

- No algorithm, but a framework for thinking about what to implement
  - Schlüter et al. (2017)
- Key points:
  - What variables in the environment does the agent perceive?
  - How do they evaluate the significance of those perceptions?
  - What aspects of the agent's state drive the options among which they need to choose?
  - How do they select an option?
  - What happens as a result?



Discussion points

- How *should* agents make decisions?
  - Are we rational?
- Should we standardize?
  - Isn't this what went wrong in economics?
- What about using LLMs?
  - Do they represent human decision-making?
  - Growing area exploring this…



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